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Global plant-symbiont organization and emergence of biogeochemical cycles resolved by evolution-based trait modelling

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evolution-based trait modelling**

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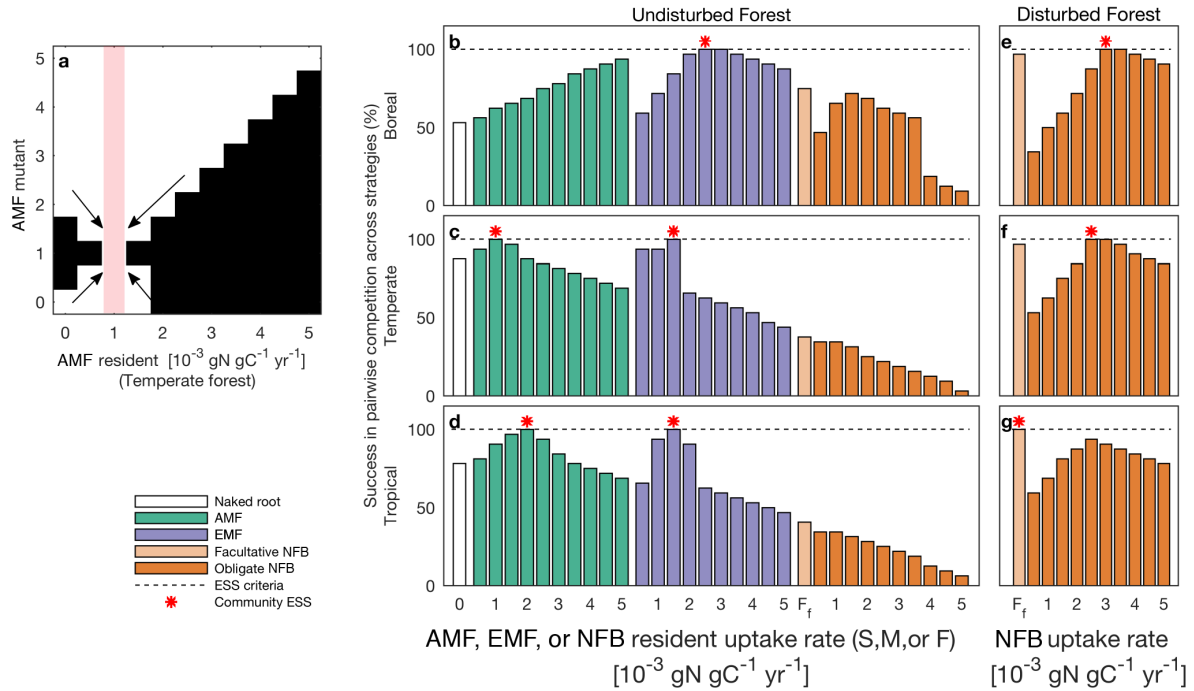
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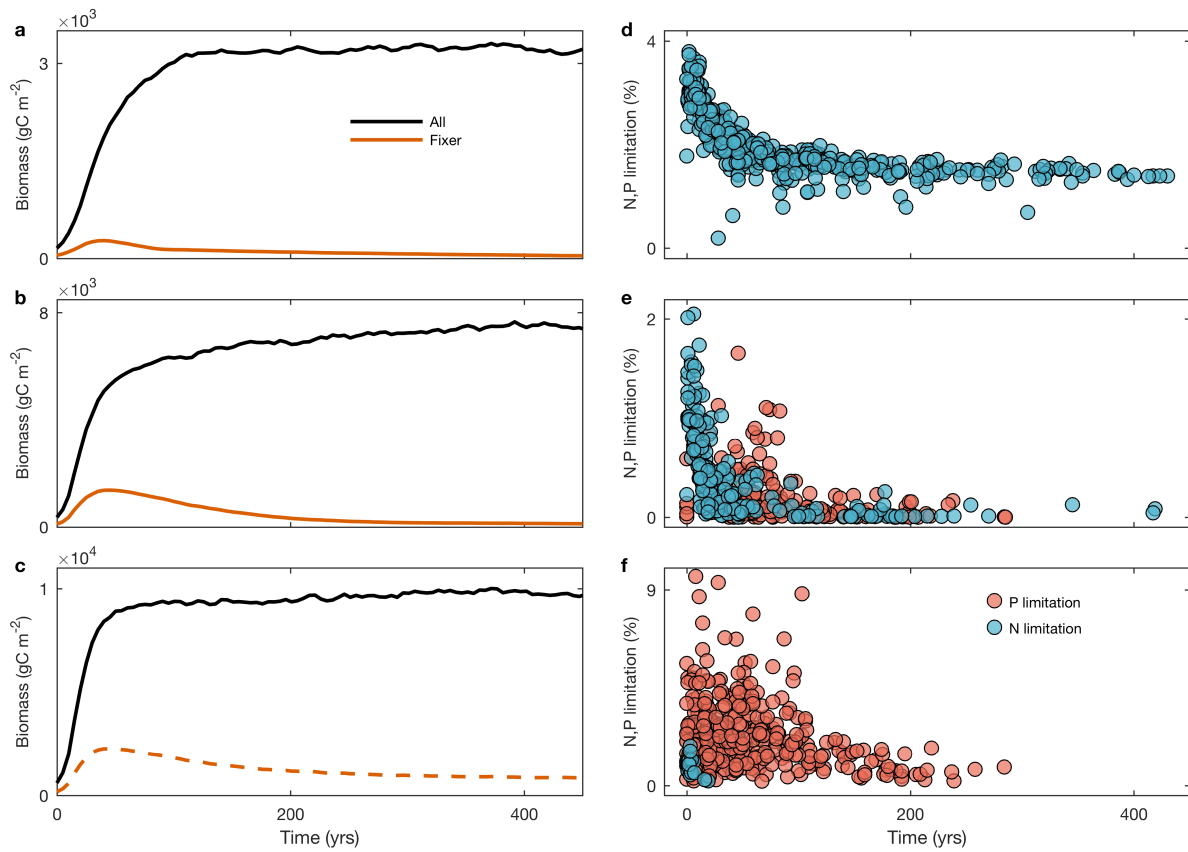
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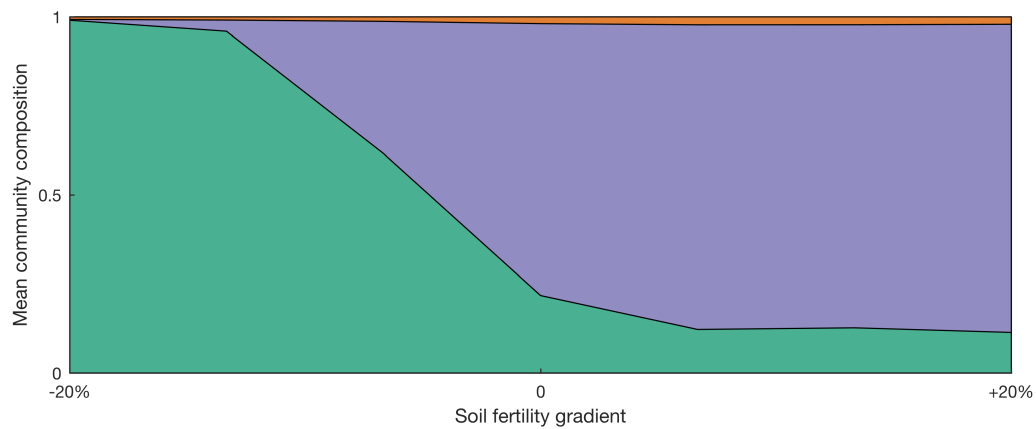
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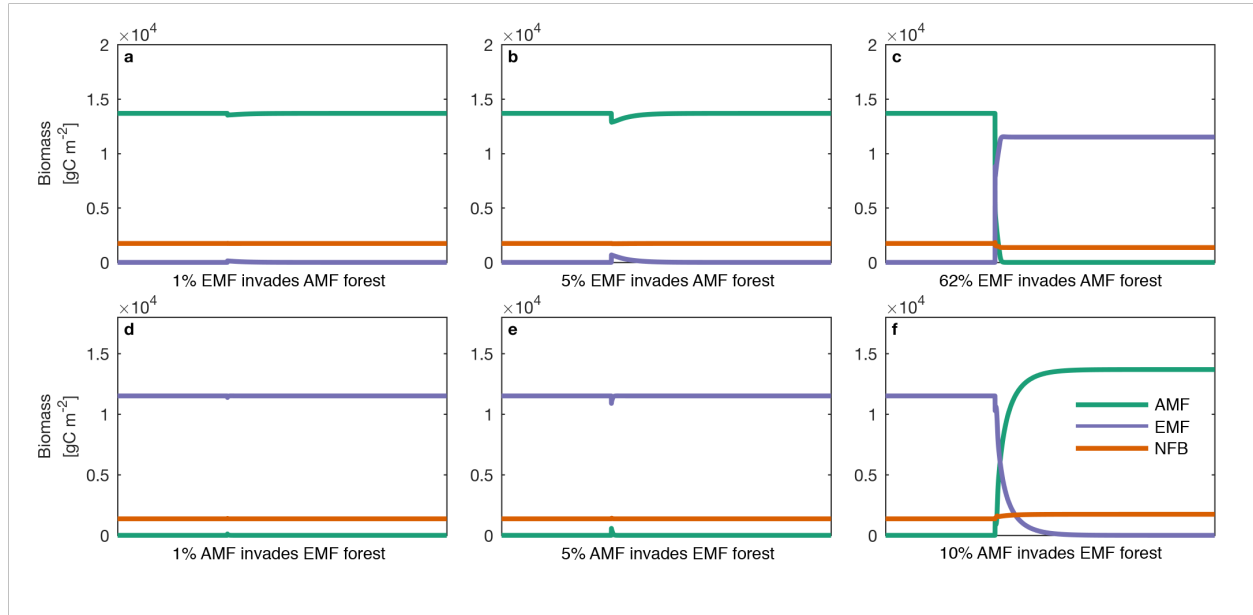
Supplementary Figure 1| Evolutionary stable strategies (ESS) of *AMF*, *EMF* and *NFB* in undisturbed and disturbed forests. **a**, A numerical pairwise invasibility matrix used to identify the ESS rate for *AMF* soil nitrogen uptake (*S*) in an undisturbed temperate forest. Black areas indicate resident-mutant combinations where the mutant eliminates the resident. The ESS is the resident rate *S* that can resist all mutant invasions (*i.e.*, 100% success in pairwise competition). Arrows show evolutionary trajectories between residents and mutants, converging on the resident ESS strategy (vertical white line). For illustration, we show only the *AMF* vs. *AMF* pairwise interactions, but to calculate the community-wide ESS we apply the analysis across all strategies. **b-d**, Resident success as a function of nutrient acquisition rate (*S*, *M*, or *F*) in undisturbed forests of each biome. Success is calculated as % of all possible mutants across all strategies resisted by the resident (*i.e.*, % white areas in a vertical column of the pairwise invisability matrix). The red star indicates the resulting community-wide ESS, defined as the strategy that can resist invasion by all possible mutants (dashed 100% line). *AMF* (green bars), *EMF* (purple bars), and *NFB* (orange, with light-orge bar identifying facultative *NFB*). The *naked root* strategy is the condition where *AMF*, *EMF*, and *NFB* uptake is zero (white bar). **e-g**, Resident success as a function of nitrogen fixation rate (*F*) in frequently disturbed forest conditions ($d=1/20\text{yr}^{-1}$) of each biome. Neither *AMF* nor *EMF* displayed community-wide ESS.



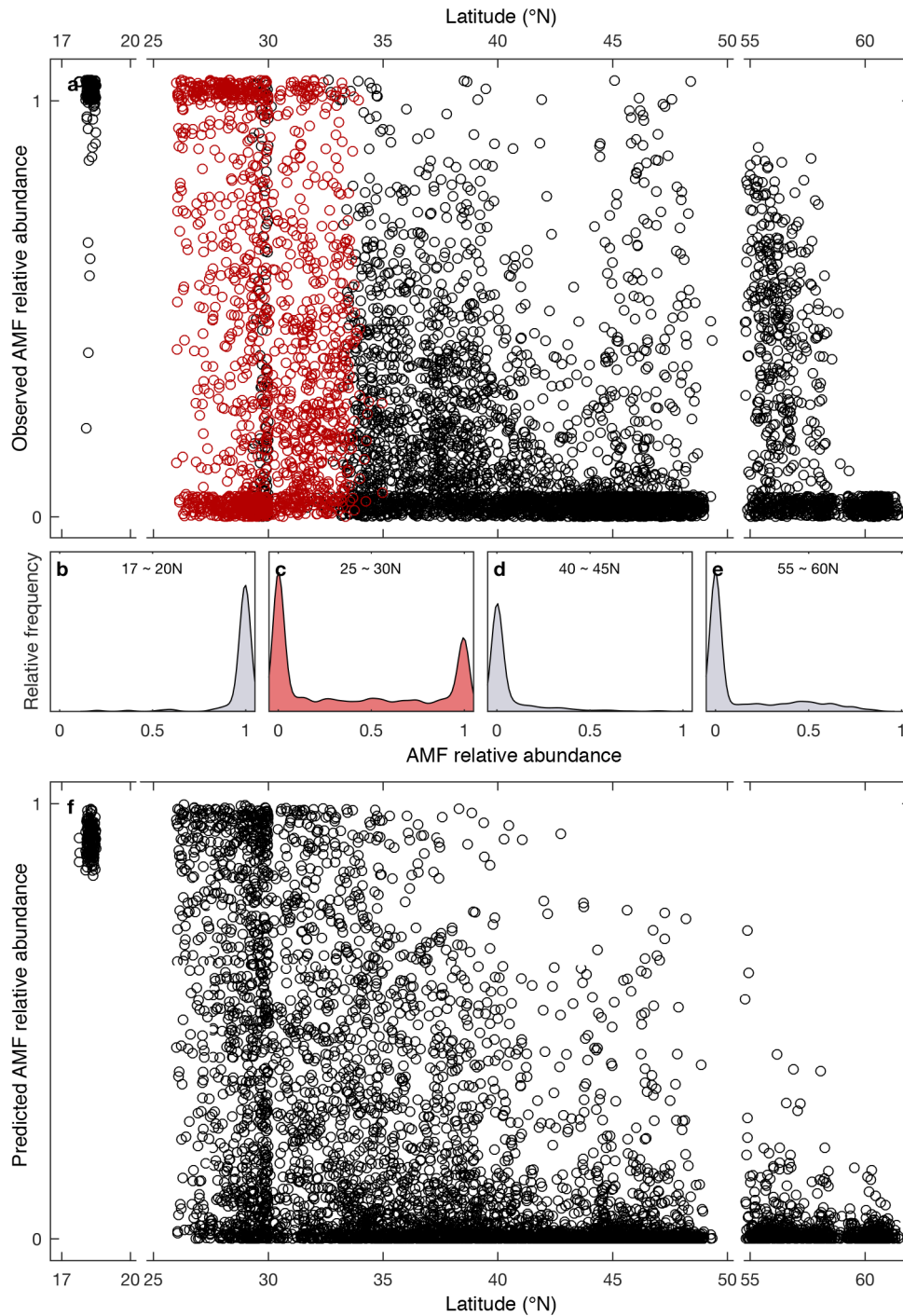
Supplementary Figure 2| Prediction of community composition and nutrient limitation across landscape succession. a-c, Landscape succession across boreal (a), unglaciated temperate (b) and tropical forest (c) biome following 95% of biomass destruction (Methods). Lines represent biomass average of 400 patches, with total community biomass in black and fixer biomass in orange. The dashed orange line denotes facultative fixer. **d-f,** Nutrient response of each forest patch in relation to the forest age across boreal (d), unglaciated temperate (e) and tropical forest (f). For all biomes, limitation was strongest in newly disturbed young forests when the supply of soil nutrients could not keep up with the demand created by rapid biomass accumulation.



Supplementary Figure 3| Change in temperate forest community composition across a soil phosphorus fertility gradient. Low soil phosphorus fertility (m_P) increased the relative abundance of *AMF* trees, while high phosphorus fertility increased the abundance of *EMF* and *NFB* trees. The *NFB* trees are known to be limited by phosphorus and thus respond to elevated phosphorus availability.



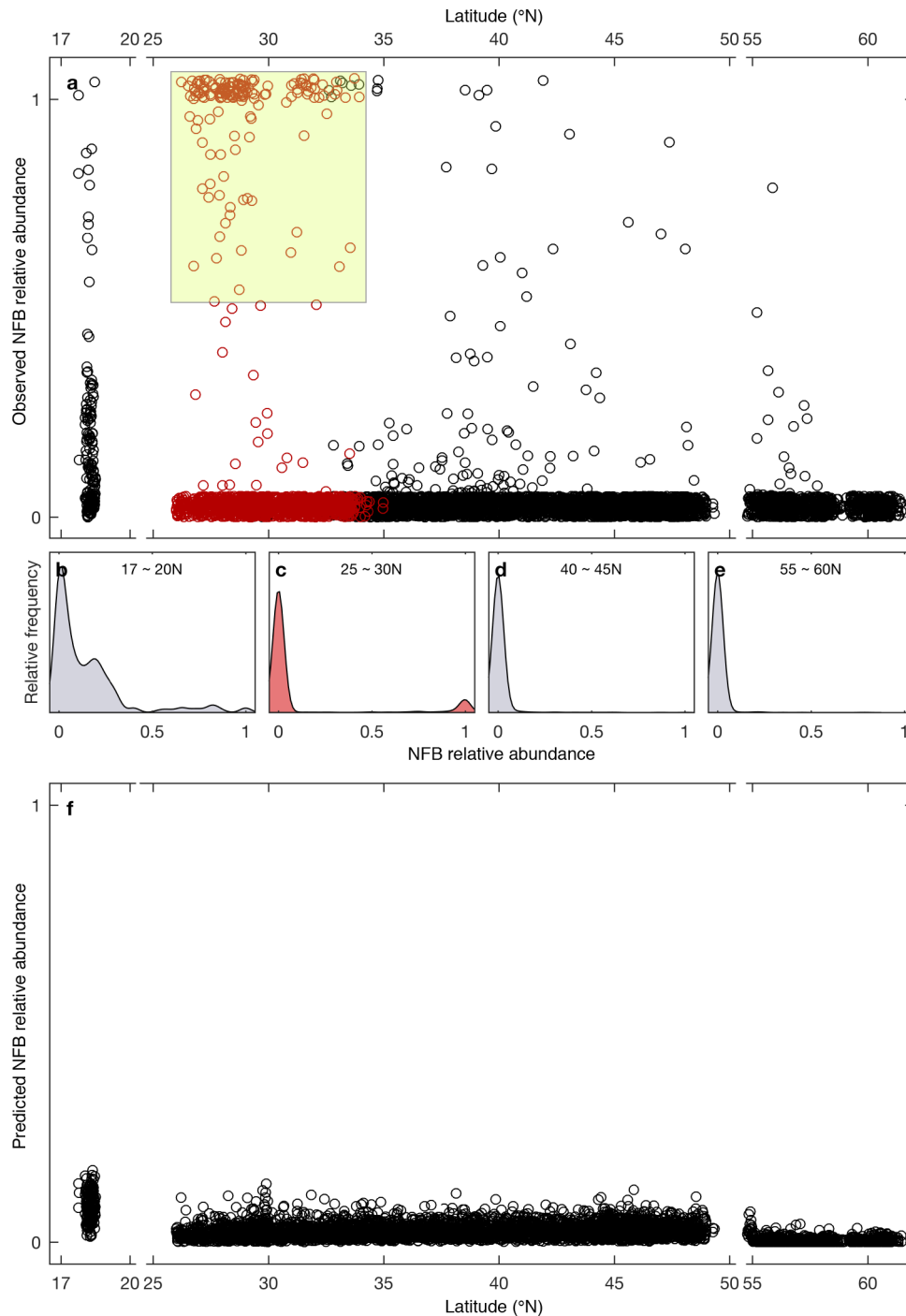
Supplementary Figure 4| Tropical forest bistability and threshold analysis. A tropical forest community (same external conditions as main text Fig.4b) with *AMF* and *NFB* trees can resist invasion by *EMF* trees that are seeded at 1% (a) and 5% (b) biomass abundance. However, the *AMF* plus *NFB* community can be invaded if *EMF* trees are seeded at ~62 % or greater abundance (c), as the resulting internal nitrogen cycle favors competition of *EMF* over *AMF* plus *NFB* trees. In contrast, a resident community of *EMF* trees can resist invasion of *AMF* trees at ~10 % initial *EMF* biomass (d-f), above which *AMF* trees exert sufficient influence on the internal nitrogen cycle so as to favor competition of *AMF* over *EMF* trees.



Supplementary Figure 5| Observed and predicted *AMF* relative abundance across FIA plots, and bistable zone in the temperate United States. a, The observed *AMF* tree basal area of each inventory plot ($n=6159$) across the FIA database (17.8-61.5 °N) shows a broad trend of decreasing *AMF* dominance with latitude. A 5% random jitter was added to the Y axis for better visibility of overlapping points. The red plots are *AMF*-*EMF* bistable (details of statistical

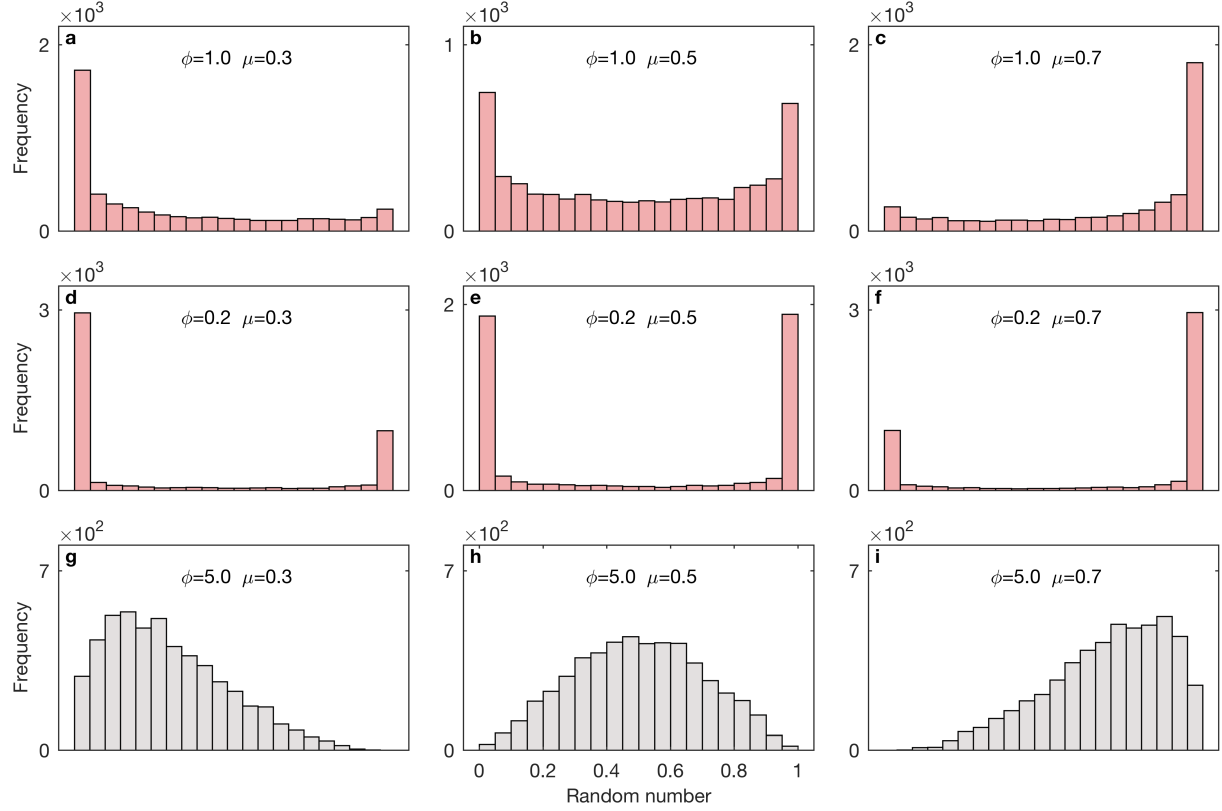
95 procedure in Methods). *AMF* is dominant in the low-latitude tropical island of Puerto Rico (**b**)
96 and is rare in the high-latitude Alaskan boreal forest (**e**). Across the temperate United States (26-
97 50 °N), *AMF* shifts from alternative stable dominance with *EMF* at the low-latitude zone (**c**) to
98 low *AMF* abundance in the high latitude zone (**d**). The area under each frequency curve equals
99 1, and we highlighted **c** in red for the bimodal distribution of *AMF* abundance in that latitude
100 band. **f**. Model prediction of *AMF* relative abundance for each plot in **a** ($n=6159$, see main text

101 Fig.5 and Methods for details).



102
 103 **Supplementary Figure 6** | Observed and predicted *NFB* relative abundance across FIA plots,
 104 and bistable zone in the temperate United States. The relative abundance of *NFB* plants is close
 105 to 0 at high latitudes (**a,d,e**) but fairly abundant in the tropical Puerto Rican forests (**a,b**). There
 106 are a number of plots with very high *NFB* abundance in the *AMF-EMF* bistable zone (**a,c**; ~240

plots >50% *NFB*). ~94% of these plots are dominated by a genus of nitrogen fixing plant—*Prosopis*—a legume shrub-tree local to the Southern US (Texas, New Mexico, Puerto Rico, and Arizona). Specifically, these plots are dominated by the native species *Prosopis glandulosa*, also known as ‘honey mesquite’. It is known to be nitrogen-fixing and arbuscular mycorrhizal and is considered among the worst invasive species outside of its home range. Our model predicts (f) very low abundance of *NFB* across FIA and agrees with the observed pattern. The yellow box highlights a zone containing ~200 high *NFB* plots that do not agree with our model production, due to the fact that these plots are from arid environments and beyond the applicability of our model. The mean aridity index of these plots is 0.37 (semi-arid to arid).



Supplementary Figure 7| Frequency distribution of 5000 random numbers drawn from beta distributions, illustrating boundary criterion for our statistical procedure in screening for bistable plots. We used a beta regression statistical approach that was developed to model proportional data⁸⁰. A beta distribution density function in its simplest form can be written as: $f(y; \mu, \phi) = cy^{\mu\phi-1}(1-y)^{(1-\mu)\phi-1}$, $0 < y < 1$, with $0 < \mu < 1$ and $\phi > 0$. μ defines the central tendency and ϕ defines the dispersion of the distribution (smaller ϕ indicate stronger dispersion). We simulated 5000 random proportional numbers (between 0 and 1) from beta distributions. **a-c** illustrates the boundary condition for our bistability analysis. A bimodal distribution with a dispersion coefficient $\phi < 1$ and population mean μ between 0.3 and 0.7 is defined as a bistable population (indicated in same red color as in main text Fig.5c). **d-f**, illustration of bistable distribution where dispersion is stronger than our boundary condition ($\phi < 1$). **g-i**, illustration of distribution that cannot be classified as bistable, where the dispersion coefficient is weaker than our boundary condition ($\phi > 1$) and the distribution is unimodal.

132 **Supplementary Table 1| Parameters used for the model**

Parameters		Units	Value
α	Aboveground fraction of plant biomass	1	0.66
$\omega_{N,P}$	C:N and C:P ratio of plant biomass	gC gN ⁻¹	50,500
c_p	Half-saturation constant for canopy photosynthesis	gC m ⁻²	3000
$\bar{v}$	Maximal uptake rate of plant-available N	gN gC ⁻¹ yr ⁻¹	0.01
$\bar{S}, \bar{M}, \bar{F}$	Maximal rate of N scavenging by <i>AMF</i> , mining by <i>EMF</i> , N fixation by <i>NFB</i>	gN gC ⁻¹ yr ⁻¹	0.005
$\hat{S}, \hat{M}$	Rate of <i>AMF</i> scavenging or <i>EMF</i> mining by nitrogen fixing plants	gN gC ⁻¹ yr ⁻¹	0.001
$\rho_{v,S,M,F}$	Translation coefficients from N uptake rate to P uptake rate	gP gN ⁻¹	0.1,0.3,0.1,0
$c_N^{v,S,M}$	Half-saturation constant for N acquisition by roots, <i>AMF</i> , and <i>EMF</i>	gN m ⁻²	0.01,0.007,800
$c_P^{v,S,M}$	Half-saturation constant for P acquisition by roots, <i>AMF</i> , and <i>EMF</i>	gP m ⁻²	0.001,0.0005,80
θ_v	Carbon cost of active uptake by absorptive roots	gC gN ⁻¹	0.5
$\theta_{S,M,F}$	Metabolic carbon cost of supporting <i>AMF</i> fungi, <i>EMF</i> fungi, and nodules	gC gN ⁻¹	2,5,10
ψ	Opportunistic carbon cost due to facultative N ₂ -fixation	gC gN ⁻¹	25
δ	Respiration rate of plant tissue	yr ⁻¹	0.05
$\mu_{S,M,F}$	Strategy specific turnover rate of plant biomass	yr ⁻¹	0.05,0.05,0.06
$\phi_{N,P}$	Hydraulic loss rate of organic nitrogen and phosphorus	yr ⁻¹	0.0005
$k_{N,P}$	Turnover rate of soil available nitrogen and phosphorus	yr ⁻¹	5
$\sigma_{S,M,F}$	Strategy specific inhibitory effect on mineralization rate	1	0,-0.36,0
τ_N	% N loss from killed biomass in disturbance event	1	0.1
v'_X	Reduction of v for maximal <i>scavenger</i> , <i>miner</i> , and <i>fixer</i>	gN gC ⁻¹ yr ⁻¹	0.0025

133 The above table summarizes all variables that do not differ across biomes, except for
 134 atmospheric nitrogen and phosphorus deposition (I), which are given together with the nutrient
 135 cycle analysis in main text Table 1 and Supplementary Table 2.

Supplementary Table 2| Emergent biogeochemical properties from landscape simulations plus biome-specific model parameters.

Variable (Units)	Boreal	Temperate	Tropical
a). Prescribed biome-specific variables			
Disturbance rate, d (yr ⁻¹)	1/100	1/70	1/50
Maximum GPP, G (t C ha ⁻¹ yr ⁻¹)	10	16	30
Phosphorus mineralization rate, m_P (yr ⁻¹)	0.003	0.011	0.025
Phosphorus deposition, I_P (kg P ha ⁻¹ yr ⁻¹)	0.62	0.62,0.55	0.55
b). Emergent C and P pools			
Plant biomass carbon (t C ha ⁻¹)	31	74	121
Soil organic carbon (t C ha ⁻¹)	156	164	76
Soil organic phosphorus (t P ha ⁻¹)	0.90	0.89	0.60
c). Emergent C and P fluxes			
GPP (t C ha ⁻¹ yr ⁻¹)	4.0	9.7	21.4
Internal phosphorus cycling (kg P ha ⁻¹ yr ⁻¹)	3.6	8.8	15.3
Total phosphorus loss (kg P ha ⁻¹ yr ⁻¹)	0.62	0.60	0.54

Each temperate biome value is the simple mean of values from the glaciated and the non-glaciated temperate biome.

Supplementary Table 3| Emergent biogeochemical properties from landscape simulation without belowground symbioses.

Variable (Units)	Boreal	Temperate	Tropical
a). Emergent C and N pools			
Plant biomass carbon (t C ha ⁻¹)	26(31)	59(74)	63(121)
Soil organic carbon (t C ha ⁻¹)	146 (156)	112 (164)	38(76)
Soil organic nitrogen (t N ha ⁻¹)	8.3(8.9)	6.0 (8.7)	3.0(5.9)
Percent fixer biomass (%)	0(<0.5)	0 (2.5)	0(8.4)
b). Emergent C and N fluxes			
GPP (t C ha ⁻¹ yr ⁻¹)	3.5 (4.0)	8.9 (9.7)	16.9(21.4)
Internal nitrogen cycling (kg N ha ⁻¹ yr ⁻¹)	29(36)	70(88)	78(153)
Nitrogen fixed by trees (kg N ha ⁻¹ yr ⁻¹)	0(0.11)	0(1.3)	0(8.5)
Total nitrogen loss (kg N ha ⁻¹ yr ⁻¹)	5.0(5.2)	5.4(7.1)	5.1(9.2)

Values presented here are the mean ecosystem properties derived from 400 landscape patches when all belowground symbioses are turned off, with values in parentheses identifying mean properties when symbioses are present (main text Table 1).

Supplementary Note 1| Cost tradeoffs for belowground strategies

We used the following cost functions for nitrogen uptake by belowground symbiotic (*NFB*, *EMF*, and *AMF*) and *naked root* strategies. We expressed the cost coefficient for both nitrogen and phosphorus as carbon cost per unit nitrogen assimilated. We used forest empirical studies to derive reasonable cost estimates for nitrogen uptake (discussed below). Forest data for phosphorus is rare, and we scaled the cost of phosphorus uptake to that of nitrogen as described in the main text. The overall per-biomass cost to the plant carbon economy follows the sequence (from highest to lowest cost): $NFB > EMF > AMF > \text{naked root}$.

1. *Nitrogen fixation (NFB)*: The carbon cost of *NFB* (θ_F) is determined by the biochemical cost of splitting the dinitrogen triple bond as well as the construction and maintenance costs of root nodules that house symbiotic *Rhizobium* or *Frankia* bacteria. Total cost estimates range between 5-12 g C per g N fixed (reviewed by Minchin and Witty 2005⁸¹; Gutschick 1981⁸²). In addition, facultative fixers can down-regulate fixation in high soil nitrogen environments and incur an additional opportunistic cost (ψ) of maintaining a root structure that is capable of both fixation and competition for soil nitrogen as discussed in ref. 5. Following ref. 5, we used $\theta_F = 10$ g C per g N acquired by fixation and $\psi = 25$ g C per g N.

2. *Ecto- and ericoid mycorrhizae (EMF)*: The carbon cost of *EMF* (θ_M) depends on the construction and maintenance of an extensive system of dense fungal hyphae² and the excretion of carbon-rich molecules that include extracellular enzymes^{1,12}. We estimated the cost from forest stand measures⁸³ of 4 tree species that possessed the *EMF* symbiosis (*Abies amabilis*, *Pinus sylvestris*, *Pinus radiata*, and *Betula nana*). We calculated the *EMF* cost coefficient as the quantity of carbon allocated to the *EMF* symbiosis divided by the estimated nitrogen flux through the *EMF* hyphae. For the carbon flux, we used field-estimates of net primary production

(NPP) and of % NPP allocated (~15%, on average) to the *EMF* symbiosis from 6 non-fertilized forest stands in 3 different studies (compiled in ref. 83). We then calculated the *EMF* nitrogen flux as the tree nitrogen biomass increment (NPP / ω , where ω is the biomass C:N ratio) multiplied by the percent of whole-plant nitrogen assimilated by the *EMF* symbiosis. Based on ^{15}N field observations⁸⁴ we conservatively assumed that *EMF* contribute ~50% of the whole-plant nitrogen flux. Finally, we assumed that the *EMF* nitrogen flux rate in these studies was at 1/3 of the maximum capacity, M (see Eq. 1b). As a result, $\theta_M = (\text{NPP} * \% \text{ NPP allocated to } EMF) / (\text{NPP} / \omega * \% \text{N}_{EMF} * 3) = 5 \text{ g C per g N}$.

3. *Arbuscular mycorrhizae (AMF)*: The carbon cost of *AMF* (θ_S) is generally considered to be significantly lower than *EMF*². *AMF* fungi deploy hyphae that are less extensive and less dense than *EMF*² that do not rely as heavily on the excretion of carbon-rich molecules or extracellular enzymes^{11,83-86}. Moreover, *AMF* plants are thought to be less dependent on the symbiosis for nitrogen uptake than *EMF* trees¹¹. We calculated the *AMF* cost coefficient as the carbon allocated to *AMF* symbiosis divided by the estimated nitrogen flux through *AMF* hyphae. For carbon, we assumed that trees allocate ~12 % of assimilated carbon dioxide (~6 % of NPP) to the *AMF* symbiosis (ref. 87 and citations therein). For nitrogen, we assumed that the *AMF* symbiosis contributes ~30% of nitrogen to net primary production, consistent with the 20-50% range observed in laboratory studies⁸⁸⁻⁹⁰. Finally, we assume that the bioavailable nitrogen concentration in the average forest soil is below the Michaelis-Menten half-saturation constant (c_S in Eq. 1b), such that the nitrogen flux through *AMF* hyphae was 1/5 of the maximum possible. As a result, $\theta_S = (\text{NPP} * \% \text{ NPP allocated to } AMF) / (\text{NPP} / \omega * \% \text{ N}_{AMF} * 5) = 2 \text{ g C per g N}$.

190 4. *Naked root*: Nitrogen uptake by bare roots includes two components: (1) free passive
191 uptake (diffusion and advection) and (2) active uptake by the root hair surface transporter that
192 requires reducing power in the form of ATP. The carbon cost of naked root uptake (θ_v) is small
193 compared to the costs associated with root construction and maintenance (δ and θ_v in Eq. 1a,
194 respectively), both of which are explicitly considered in our model. As a result, we used the
195 value $\theta_v = 0.5$ g C per g N.

196 Despite the lower carbon cost, the naked root strategy is less competitive than *AMF* at low
197 bioavailable nutrient levels since it has lower affinity (*i.e.*, higher Michaelis-Menten K_m) for
198 dissolved nitrogen. At high bioavailable nitrogen, however, the naked root strategy has a
199 significantly higher maximum uptake rate (*i.e.*, higher Michaelis-Menten V_{max}) than *AMF* (SI
200 Table 1).

201 **Supplementary Note 2| Sensitivity analysis of cost coefficients**

202 Our analysis is robust to changing the assumptions about the cost of belowground symbioses.
203 Most generally, our results emerge as a function of the relative sequence in cost of each
204 symbiosis (from highest to lowest cost): $NFB > EMF > AMF > \text{naked root}$.

205 First, we systematically increased or decreased the cost coefficients by the same factor, while
206 keeping the relative sequence constant. Our results (supplementary Fig.1 and main text) were
207 qualitatively robust to changes in the cost coefficients of up to at least $\pm 30\%$. Quantitatively,
208 however, higher cost coefficients shifted the community-level ESS of each strategy towards
209 lower symbiotic investment (X) and vice versa.

210 Second, our model is also reasonably robust to changes in individual cost coefficients, as
211 long as the overall between-symbiosis cost structure is maintained. The *naked root* and *NFB*
212 strategies are both largely insensitive to cost variations up to $\pm 30\%$ or more. The *naked root*

strategy is robust because the cost coefficient is much smaller than all of the symbiotic strategies. The *NFB* strategy is robust because of its unique dependence on disturbance: it emerges as the community-level ESS in disturbed forests with low soil nitrogen, even as θ_F increases. *AMF* and *EMF* are more sensitive to changes in cost coefficients, but also in this case the sensitivity is quantitative rather than qualitative. For example, increased *EMF* cost translates into *EMF* trees becoming less competitive (relative to *AMF*) in temperate forests and in low nitrogen soils. Conversely, increased *AMF* cost translates into *AMF* trees becoming less competitive in temperate and tropical forests, or in high nutrient soils. The overall pattern therefore is a shift of the balance between *AMF* and *EMF* symbioses across latitude and soil nutrient conditions in response to variations in their relative cost.

Third, even when variations in *AMF* or *EMF* cost coefficients caused changes in the community-level ESS analysis (Fig. 2), these were generally small. For example, increasing θ_M by 20% in tropical forests caused the percent success of the best performing mutant to decline slightly (from 100% to 94%). In turn, these differences had relatively muted effects on the landscape dynamics analysis (Fig. 3a) or the succession analysis (Supplementary Fig. 3a-c) because in both analyses, we selected the best performing mutant from each strategy and the percent success of this mutant (relative to all other strategies) was relatively insensitive to variation in θ_M . Moreover, the heterogeneous conditions of soils and forest succession caused by stochastic patch disturbance further buffered against dramatic changes in the community composition in our landscape analysis.

Supplementary Note 3| Biological basis for the root uptake tradeoff and sensitivity analysis

Plants inevitably sacrifice the *naked root* area available for nutrient uptake when colonized by various symbionts. To capture this essential “uptake tradeoff” we used a linear⁵ function (Eq. 4,

Methods) in which the uptake capacity (v) of the *naked root* absorptive system decreased linearly with increased investment in a symbiosis (X). This tradeoff is apparent in studies across the symbioses considered in this study: First, *AMF* colonization has been shown⁹¹ to reduce root length by 31% and root hair density by 43%. *AMF* has also been shown to inhibit *naked root* nutrient uptake, by funneling inorganic nutrients through *AMF* hyphae⁹². Second, *EMF* generally forms a fungal hyphae sheath that encapsulates the rootlets², arresting the cell division of the ensheathed roots⁹³, and excreting chemicals (*e.g.*, hypaphorine) that suppress the development of absorptive root hairs⁹⁴. Third, *NFB* nodules and lateral roots mutually exclude each other by competing for the same active developmental sites⁹⁵. The emergence of nodules thus means the reduction of lateral roots, which could have branched into further orders of absorptive roots that aid in nutrient uptake. Over evolutionary time, plants may be able to evolve more efficient, more finely branched rooting systems— that depend less on symbiotic relationships¹⁵—however, for a given rooting architecture, the per biomass uptake surface area will monotonically decrease with increasing symbiotic load.

We examined the sensitivity of our results to the assumption of a linear uptake tradeoff (Eq.4) using a general non-linear function:

$$v(X) = ae^{\epsilon X} + b \quad (S1)$$

where a , b are constants calculated based on v'_X , $\bar{v}$ and $\bar{X}$ to ensure that Eq.S1 and Eq.4 describe the same symbiont-free state and the same maximal symbiont state. The absolute value of ϵ controls the strength of curvature of the nonlinear function, while the sign of ϵ controls the direction of the relationship (negative sign denotes convex downward function form, positive concave upward).

Quantitatively, our ESS results remain virtually the same for ϵ ranging from -10 to 10 (mild nonlinearity). For strongly non-linear functions, the ESS results change but remain qualitatively robust: For example, at $\epsilon = -150$ our best performing *AMF* in temperate and tropical forest increased by $\sim 10\%$ in their scavenging rate. At $\epsilon = 150$, our best performing *AMF* in the tropics dropped its scavenging rate by $\sim 10\%$. These differences have significant but relatively minor effects on our model predictions of community composition across latitude or successional age. We conclude that non-linearity has relatively minor impact on our overall inferences about broad patterns in community composition.

Supplementary Note 4| Biological basis for the plant-soil-nutrient feedback

Our model captures two key mechanisms in which belowground strategies can influence the local plant-soil nutrient cycle. First, nitrogen-fixing trees can bring in new atmospheric-derived nitrogen to the ecosystem, at rates that influence the fixer vs. non-fixer competition over successional time (*e.g.*, ref. 5). Our model includes this mechanism in the form of the fixation rate (F) and competition between strategies in our landscape dynamics analysis. Moreover, higher mortality and herbivory rates cause nitrogen-fixing trees to turn over faster than non-fixing trees in nature^{10,96}. Based on empirical studies⁹⁷⁻⁹⁹ we conservatively assumed 20 % higher biomass turnover (μ) in fixers than non-fixers. Second, forests dominated by *EMF* trees display on average $\sim 36\%$ slower mineralization rate (m) than *AMF* forests across biomes worldwide⁴⁷. This effect results from the combination of two mechanisms: (1) *EMF* trees on average produce recalcitrant litter (*e.g.*, high C:N ratios and high lignin content) that causes both slower rates of decomposition and mineralization⁴⁷; and (2) *EMF* trees can suppress the activity of saprotrophic soil microbes by excreting antagonistic secondary compounds or by generating conditions of strong non-symmetrical competition between the *EMF* fungi and the local

microbial community^{100,101}. In our model, we captured both effects in a single term (σ_X) designed to correct m for the *EMF* inhibitory effect (*i.e.*, $\sigma_M = -0.36$ and $\sigma_S = \sigma_F = 0$).

We evaluated whether our results were sensitive to separating the inhibition effect into separate litter vs. fungal (or “Gadgil”) effects. As illustrated in the below equation, we allowed for a direct litter effect (σ_{LX}) on m , but considered that the fungal effect (σ_{GX}) on m scaled in proportion to the relative biomass of trees from each strategy in a given forest patch.

$$\sigma_X = \sigma_{LX} + \frac{\sum \sigma_{GX} B_X}{\sum B_X} \quad (S2)$$

This formulation captures the essential mechanisms and dynamics proposed for both effects. The partitioning did not, however, have any significant influence on our results. We thus concluded that the single term σ_X is sufficient to capture the essence of the inhibitory dynamics.

Supplementary Note 5| External disturbance in the landscape patch simulations

Old-growth forest is subject to constant external disturbance (wind-throw, landslide, fire, etc.), resulting in the turnover of forest patches at the landscape level¹⁰². A global analysis²⁶ shows that tropical forests turn over at a rate of $\sim 1/44 \text{ yr}^{-1}$ for rich soil sites and $\sim 1/71 \text{ yr}^{-1}$ for the poor sites, whereas temperate forests turn over at a rate between $\sim 1/58 \text{ yr}^{-1}$ to $\sim 1/97 \text{ yr}^{-1}$, and pure conifer/boreal forests turn over at the slower rate of $1/130 \text{ yr}^{-1}$. While this global dataset might not accurately represent fire in boreal biomes, regional paleorecords¹⁰³ indicate a boreal forest turnover rate with respect to fire at $\sim 111 \text{ yr}^{-1}$ over the past 3,000 years. As a result, we used the following latitudinal disturbance gradient: Tropical forest at $1/50 \text{ yr}^{-1}$, temperate forest at $1/70 \text{ yr}^{-1}$, and boreal forest at $1/100 \text{ yr}^{-1}$. The global pattern presented in Fig.3a was robust to variation in this disturbance gradient by $\pm 20\%$ or more: Tropical forests remained co-dominated by *AMF* and *NFB*, boreal forests by *EMF* with few *NFB* trees, while temperate forests displayed a transitional status with disturbance favoring *AMF* and *NFB* over *EMF*. Higher disturbance

rates would favor the abundance of *AMF* and *NFB* in a community, while lower disturbance rates favor *EMF*.

In addition, our landscape analysis was robust to varying the intensity by which a disturbance event impacts the nitrogen cycle. Changing the nitrogen loss factor for each disturbance (τ in Eq. 2) by $\pm 50\%$ gives the same qualitative results as shown in Fig. 3a. High intensity disturbance events (*e.g.*, exceptionally hot wild fires) would cause higher nitrogen losses from the ecosystem, and thus favor the presence of nitrogen fixers in a community.

Supplementary Note 6| Soil fertility and community composition

Forest biomes extend across diverse conditions of soil fertility, generated by variation in parent material, topography, and pedogenic development. We used the nutrient mineralization rate constant (m) as a proxy for variations in soil fertility caused by these external factors. (In addition, local soil fertility is influenced by the internal ecosystem feedback between plants and soil, as described above.) Our analyses showed that the forest community composition was sensitive to variation in m in both temperate and tropical forests, but that boreal forests remained dominated by *EMF* across a reasonable range in m . For example, in the temperate biome, the forest community composition changed as a function of m_P in a predictable pattern (Supplementary Fig. 3): *NFB* and *EMF* trees were favored as the rate of bioavailable phosphorus cycling increased while *AMF* trees were favored as the rate declined. This trend broadly agrees with the empirical understanding that *AMF* trees are superior in acquiring phosphorus² and that nitrogen fixing trees are often limited by phosphorus²⁹⁻³¹.

In the tropical biome (main text Fig. 4a), *AMF* remained the sole dominant strategy at both average and high (+20% increase in m) soil fertility levels. At low soil fertility, however, *AMF*

or *EMF* alternatively dominated individual patches. We discuss this alternate dominance in the following supplementary note 7, on tropical bistability.

Supplementary Note 7| Tropical *AMF-EMF* bistability and founder effect

Our findings (main text Figs. 2c and Fig. 4) are consistent with the empirical observation^{13,21,72,73,104} that, on poor soil substrates, tropical forests can be dominated by either *AMF* or *EMF* tree species. We analyzed the ability of *AMF* trees to invade a community dominated by *EMF* trees and vice versa. Our results show that, when established on the same nutrient-poor soil, both strategies could resist the invasion of populations of the alternative strategy (Supplementary Fig.4).

We further examined whether the final equilibrium composition depended on initial conditions of community establishment. The absence of such initial dependence (hereafter “founder effect”) would imply that the final equilibrium community deterministically results from the external variables. In contrast, the presence of such a founder effect would imply that the final forest community also depends upon the very biotic community that makes up the internal components of the ecosystem. This effect therefore offers a mechanism by which a plant community can generate the conditions of nutrient cycling and soil fertility that competitively favors itself – thus embedding Darwinian selection at the scale of individual trees within the broader conditions created by the ecosystem plant-soil nutrient cycle.

A bifurcation analysis revealed that the final community composition (the two large circles in main text Fig.4b) was strongly dependent on the initial abundance of *EMF* vs. *AMF* in the founder population (small circles in main text Fig.4b). Since *AMF* and *EMF* trees establish distinctive plant-soil nutrient economies, the early dominance of either population will create the biogeochemical condition that favors their further population increase. This positive feedback

will eventually lead to the divergence of communities into two alternative stable states, despite the fact that the initial abiotic conditions were the same.

Supplementary Note 8| Temperature coefficient for the plot-level simulation

To derive the plot-level mineralization rate (m) and disturbance rate (d), we allowed both m and d to vary based on the plot specific temperature. As described in the equation below, for each rate value we approximated the plot-level value (R_2) based on a Q_{10} function:

$$Q_{10} = \left(\frac{R_2}{R_1}\right)^{10/(T_2-T_1)} \quad (S3)$$

where T_2 the plot-level mean annual temperature, R_1 denotes biome-specific rate values (m and d in Table1), and T_1 the biome-specific reference temperature. We prescribed the biome reference temperature as 5 degree for boreal, 12 for temperate and 26 for tropical, each derived from the mean temperature of low-elevation Gentry plots (<300m).

Supplementary Note 9| Differences in dynamics resulting from inclusion of phosphorus

In addition to the results reported in our main text, we also evaluated a version of our model that did not explicitly consider phosphorus. This limitation of our analyses was consistent with the great majority of land biogeochemical models, which so far have been limited to the consideration of carbon and nitrogen dynamics. Thus, we had the opportunity to compare two structural versions of the same model – carbon plus nitrogen vs. carbon plus nitrogen plus phosphorus – to understand the role of inclusion of phosphorus in generating the emergent behavior of interest.

The inclusion of phosphorus allowed for the following improvements in our predictions: (1) It allowed us to predict the occurrence of phosphorus limitation within and across biomes. Encouragingly, our model re-creates biome-scale empirical patterns of phosphorus limitation, as well as differences in the propensity of phosphorus vs. nitrogen limitation within biomes

(Figs.3c-f). (2) The empirically observed feedback between soil phosphorus and nitrogen fixation in trees emerged naturally in our model, within biomes and across successional stages (forest patches or successional stages with phosphorus limitation also limited the rate of nitrogen fixation). This result implies a general interaction of the phosphorus and nitrogen cycle within and across biomes, mediated by plants and their symbionts. (3) Our predictions and mechanistic understanding of bi-stability greatly improved when we considered phosphorus in the model. Bi-stable dynamics emerged in a region of conditions where the inorganic versus organic nutrient economies are “just right” – that is a balance of phosphorus and nitrogen that allows either *AMF* or *EMF* to establish in a landscape, but where each strategy excludes the other once established. As a result, inclusion of phosphorus allowed for the emergence of highly complex statistical distributions of *AMF* and *EMF* symbioses within- and across-biomes, including the shift from *AMF* to *EMF* dominance and the intermediate bi-stable conditions observed in >200,000 FIA plots across North America (Fig.5).

Supplementary Note 10| Invasibility analysis

We numerically evaluated competition between all paired combinations between a resident (*i.e.*, a specific trait value X) and any mutant strategy (Methods). We examined whether the introduction of a mutant population (B^m at 1% of the resident biomass) at equilibrium could invade the resident population (B^r), given the conditions established by the resident:

$$\frac{dB^r}{dt} = (\min(g_C^r, g_N^r, g_P^r) - \mu - d_j) B^r \quad (S4)$$

$$\frac{dB^m}{dt} = (\min(g_C^m, g_N^m, g_P^m) - \mu - d_j) B^m \quad (S5)$$

where both the mutant and the resident populations compete for aboveground (g_C^r, g_C^m) and belowground resources (g_N^r, g_N^m) (Eq.S6-9):

$$g_C^r = \frac{\alpha G_j}{\alpha(B^r + B^m) + c_G} - \delta - (1 - \alpha)(\theta_S S^r + \theta_M M^r + (\theta_F + \psi)F^r + \theta_v v^r(X)) \quad (S6)$$

$$g_C^m = \frac{\alpha G_j}{\alpha(B^r + B^m) + c_G} - \delta - (1 - \alpha)(\theta_S S^m + \theta_M M^m + (\theta_F + \psi)F^m + \theta_v v^m(X)) \quad (S7)$$

$$g_i^r = \omega_i(1 - \alpha) \left(\frac{S_i^r A_i}{A_i + c_i^S} + \frac{M_i^r O_i}{O_i + c_i^M} + F_N^r + \frac{v_i^r(X) A_i}{A_i + c_i^v} \right) \quad (S8)$$

$$g_i^m = \omega_i(1 - \alpha) \left(\frac{S_i^m A_i}{A_i + c_i^S} + \frac{M_i^m O_i}{O_i + c_i^M} + F_N^m + \frac{v_i^m(X) A_i}{A_i + c_i^v} \right) \quad (S9)$$

We tracked the organic matter that derives from the resident and mutant population separately (Methods), with both the resident nutrient pool (O_i^r) and mutant nutrient pool (O_i^m) accessible to the rooting system of both resident and mutant populations (Eq.S10-11):

$$\frac{dO_i^r}{dt} = (\mu_X + d_j - \tau_i d_j) \frac{B^r}{\omega_i} - \left(m_j(1 + \sigma_X) + \phi + \frac{(1 - \alpha)(M_i^r B^r + M_i^m B^m)}{O_i + c_i^M} \right) O_i^r \quad (S10)$$

$$\frac{dO_i^m}{dt} = (\mu_X + d_j - \tau_i d_j) \frac{B^m}{\omega_i} - \left(m_j(1 + \sigma_X) + \phi + \frac{(1 - \alpha)(M_i^r B^r + M_i^m B^m)}{O_i + c_i^M} \right) O_i^m \quad (S11)$$

The available nutrient pool combines the mineralized nutrients from separate organic nutrient pools, and can be accessed by the rooting system of both resident and the mutant populations (Eq.S12):

$$\frac{dA_i}{dt} = I_i + m_j \left((1 + \sigma_X^r) O_i^r + (1 + \sigma_X^m) O_i^m \right) - \left(k_i + \frac{S_i^r B^r + S_i^m B^m}{A_i + c_i^S} + \frac{v_i^r B^r + v_i^m B^m}{A_i + c_i^v} \right) (1 - \alpha) A_i \quad (S12)$$

Supplementary Video 1| Tropical forest community successional dynamics. Visual demonstration of successional dynamics (1000yrs) from a nutrient-poor tropical forest using a landscape of 15*15 patches. Each patch is represented by 3 trees, *AMF* in green, *EMF* in purple, and *NFB* in orange color, with the tree height and crown radius proportional to the total biomass

411 of each symbiotic group in each patch. The landscape recovered from 99% destruction of its
412 equilibrium biomass, with *NFB* accounting for 10% of starting biomass and *AMF* and *EMF* each
413 accounting for 45%. *NFB* dominated early succession, but the strategy was increasingly
414 outcompeted by *AMF* as succession proceeded. In late succession, *EMF* became abundant and
415 coexisted with *AMF* at the landscape scale as forest biomass again approached equilibrium.
416 Throughout succession, stochastic disturbance occurred at the scale of individual patches, and
417 contributed to the presence of *NFB* trees in the landscape.

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